

GÖKHAN TUTKU ÇAY

Senior Game Developer & Systems Engineer (Unity / Gameplay / F2P Systems)

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GitHub: github.com/gkhanC | **Live Portfolio:** Portfolio Repository

EXECUTIVE SUMMARY

Senior Game Developer and Systems Designer with **5+ years of hands-on production experience** in Unity (C#), live game deployment, and real-time multiplayer systems. Shipped commercial products across both **Steam (*Fuu*)** and **Google Play (*WordQuest*, 1K+ installs with IAP/Ad monetization)**, alongside 50+ rapid prototypes. Deeply proficient in bridging mathematical game design with high-performance engineering—translating combat formulas, difficulty progression curves, and in-game economy loops (sources/sinks) into scalable, 60 FPS architectures.

TECHNICAL SKILLS & SYSTEMS ARCHITECTURE

Core Gameplay & AI Frameworks
Physics-based Character Controllers, Combat & Ability Systems, Inventory & Loot Logic, Finite State Machines (FSM), Behavior Trees (AI), Unity NavMesh, Trigger/Obstacle Puzzles.

Game Economy & Progression Balancing
Mathematical modeling of currency sinks/sources, loot drop probability tables, upgrade cost balancing, difficulty escalation curves, Excel/Sheets numerical simulations.

Profiling & Mobile Optimization
Unity Profiler, Garbage Collection reduction, Draw Call Batching, Memory Footprint Optimization, Shader & Asset Tuning, Stable 60 FPS targeting on low/mid-tier mobile hardware.

Data-Driven & Modular Architecture
ScriptableObject Architecture, Entity Component System (ECS), Data-Oriented Design (DOD), SOLID, Design Patterns (Factory, Observer, State, Command, MVC), Custom Unity Inspectors.

Multiplayer & Real-Time Networking
Unity Photon (Fusion/PUN), Client-side Prediction, Rollback & Lag Compensation, WebSocket, Client/Server sync, Authoritative Server architecture, 1v1 & 3v3 arenas.

Monetization, Backend & Tools
F2P Mechanics, In-App Purchases (IAP), Rewarded Video Integration, RESTful Web APIs, SQL (Server/TSQL), Steamworks API, Jenkins CI/CD, Git Version Control.

PROFESSIONAL EXPERIENCE

Independent Game Developer & Technical Lead | **Self-Employed / Remote** — 2024 – Present

- **Commercial Steam Release (Fuu):** Shipped a 2D precision platformer runner featuring custom double-jump, wall-jump, dash/slide physics, and logic-based trigger puzzles. Managed entire release lifecycle, QA, Steamworks API integration, and post-launch updates.
- **50+ Rapid Market Prototypes:** Evaluated global F2P and indie mechanic trends to develop 50+ modular prototypes. Accelerated iteration speed by **30%** via decoupled ScriptableObject templates and procedural framework design.
- **Combat & Systems Balancing:** Formulated mathematical models for damage escalation, dynamic difficulty curves, and player ability cooldowns, balancing gameplay mechanics directly inside Unity custom tools.
- **Cross-Platform Optimization:** Enforced zero-allocation GC patterns and batch rendering, guaranteeing a sustained 60 FPS on diverse PC and mobile screen environments.

Senior Game Developer | **Gamefacto** — 2021 – 2024

- **Scalable Gameplay Systems:** Built core combat, progression, and level interaction architectures for production titles using event-driven, decoupled C# modules.
- **Performance Tuning & Bug Remediation:** Systematically investigated and resolved high-priority bottlenecks using Unity Profiler, driving a **40% drop in regression defect recurrence** before live builds.
- **Custom Tooling for Designers:** Programmed specialized Unity Editor tools and balancing windows, enabling game designers to configure character parameters, enemy stats, and drop tables without modifying code.
- **Engineering Mentorship:** Mentored junior developers in architectural best practices, clean code standards, and memory management conventions.

PROFESSIONAL EXPERIENCE (CONTINUED)

Team Leader & Technical Project Manager | **Dark Zone** 2021 – 2022

- **Engineering Team Coordination:** Led an 8-developer production team, facilitating sprint planning, technical backlog breakdown, and daily Agile scrums to meet strict delivery milestones.
- **Design-to-Code Alignment:** Collaborated directly with product and design leads to validate technical feasibility, ensuring mechanics aligned 100% with gameplay and monetization goals.
- **Deployment Stability:** Supervised release pipelines and risk assessments, managing incident resolutions to maintain high uptime during multi-client test rollouts.

Game Developer | **Satori Game / Blue Potion** 2020 – 2021

- **F2P Mobile Publishing (WordQuest - Google Play):** Developed and launched a live mobile word/trivia game; implemented core mechanics across 100+ categories, UI systems, and persistent player progression.
- **In-Game Economy & Monetization:** Integrated In-App Purchases (IAP), rewarded ads, randomized item loops, and virtual currency economies designed to boost player retention and ARPDAU.
- **Iterative Prototyping:** Built rapid hyper-casual and hybrid-casual loops for publisher testing, analyzing retention metrics and drop-off points to optimize early level progression.

VERIFIED IMPLEMENTATIONS & TECHNICAL PORTFOLIO

1. Fuu — Commercial 2D Precision Platformer LIVE ON STEAM store.steampowered.com/app/4081480 ↗

Engineered player controller physics (double jump, wall cling/jump, slide/dash mechanics) and dynamic trigger traps. Built full save/load serialization, input rebinding, and Steamworks integration.

2. WordQuest — F2P Mobile Casual/Puzzle Game GOOGLE PLAY (1K+ INSTALLS) play.google.com/store/apps/WordQuest ↗

Designed and deployed complete F2P gameplay systems including hundreds of data-driven word categories, in-game shop, IAP economy flow, rewarded video ads, and client-side data safety compliance.

3. Modular Combat & Enemy AI Framework [ScriptableObjects / FSM] github.com/gkhanC/Portfolyo ↗

Architected a plug-and-play combat system enabling designers to balance attack combos, damage multipliers, cooldowns, and hitbox timing without code changes. Features modular enemy state machines.

4. Real-Time Multiplayer Networking Sandbox (Flameborn) [Photon / Netcode] github.com/gkhanC/Portfolyo/flameborn ↗

Real-time competitive combat arena implementing client-side prediction, rollback synchronization, authoritative hit verification, and latency compensation over UDP.

EDUCATION & ANALYTICAL BACKGROUND

Gaziantep University — Undergraduate Studies Focus: Analytical Logic, Syntax & Formal Systems

Studied formal logic, structural linguistics, and abstract symbolic relationships. Successfully translated this foundation into systems engineering, state machine modeling, and gameplay progression algorithms.

LANGUAGES & AVAILABILITY

English: Professional Working Proficiency (Technical Documentation, Agile Syncs)

Turkish: Native

Status: Immediate Remote/Hybrid Availability